

A Parity Lower Bound for Twofold Radius-One Coverings of the Binary Hypercube

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Abstract

A binary code is a twofold covering of radius one if every word of the hypercube lies within Hamming distance one of at least two codewords. For every even length we prove a lower bound on the minimum size of such a code by a parity observation taken at a codeword rather than at an arbitrary word: in a code without repeated words the ball of a codeword carries a unique diagonal term, whose parity forces an excess whenever the length is even. After taking ceilings, the bound exceeds the general lower bound of Krotov and Potapov for every even length from six onwards. It raises the tabulated lower bounds at the five lengths eight, ten, twelve, fourteen and sixteen, and at length six it supplies an elementary lower-bound proof which, with the known twenty-word construction, recovers the exact value. The argument extends to every feasible even multiplicity; for the underlying real-valued bounds we determine exactly when the resulting expression exceeds the Krotov–Potapov expression, noting that a strict gain need not survive the ceiling.

Keywords: covering code; multiple covering; twofold covering; double domination; covering excess; hypercube; lower bound.

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1. Introduction

Let $\mathbb{F}_2^n$ be the binary Hamming space and let $B(x) = \{y : d(x, y) \leq 1\}$ be the closed ball of radius one, so $|B(x)| = n + 1$. A set $C \subseteq \mathbb{F}_2^n$ is a μ -fold covering of radius one if $|B(x) \cap C| \geq \mu$ for every $x \in \mathbb{F}_2^n$, and $K(n, 1, \mu)$ denotes the least cardinality of such a set. Codewords are distinct: C is a set and not a multiset. Allowing repetitions gives a different quantity, treated separately in [1], and the hypothesis that C is a set is used essentially below. We write $K(n, 1, 2)$ for the twofold case. Equivalently, such a set is a μ -tuple dominating set in the n -dimensional hypercube graph, and a double dominating set when $\mu = 2$.

The standard tool for lower bounds of this kind is the covering excess method, extended to multiple coverings by Härmäläinen, Honkala, Kaikkonen and Litsyn [1]. Their Theorem 6 improves on the sphere covering bound by an additive correction term ε which, at radius one, vanishes exactly when $\mu(n + 1)$ is even. For even n that is exactly when μ is even, so at $\mu = 2$ and n even their bound reduces to the sphere covering bound; see Section 7.

Our observation is that a *different* parity is available, and that it fires precisely where theirs vanishes. Summing the coverage function over the ball of a *codeword* rather than of an arbitrary word produces a sum with a unique diagonal term $n + 1$, hence a sum that is odd whenever n is even, forcing an excess in every codeword ball. The resulting bound is Theorem 1.1.

Theorem 1.1. *Let $n \geq 2$ be even and let $C \subseteq \mathbb{F}_2^n$ satisfy $|B(x) \cap C| \geq 2$ for every $x \in \mathbb{F}_2^n$. Then*

$$|C| \geq \frac{3 \cdot 2^{n+1}}{3n + 2}, \quad \text{and hence} \quad K(n, 1, 2) \geq \left\lceil \frac{3 \cdot 2^{n+1}}{3n + 2} \right\rceil.$$

The proof occupies half a page and uses nothing beyond double counting. Section 4 shows that after taking ceilings the bound exceeds the general lower bound of Krotov and Potapov [2] at every even $n \geq 6$; Section 5 records the consequences at small n ; and Section 6 carries the argument to every feasible even multiplicity and classifies when the underlying real-valued bound exceeds the Krotov–Potapov bound.

2. Notation

For $C \subseteq \mathbb{F}_2^n$ put

$$c(y) = |B(y) \cap C|, \quad g(y) = c(y) - 2, \quad E = \sum_{y \in \mathbb{F}_2^n} g(y).$$

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If C is a twofold covering then $g \geq 0$ pointwise. Counting the incidences (y, z) with $z \in C$ and $y \in B(z)$ in two ways gives $\sum_y c(y) = (n+1)|C|$, whence

$$E = (n+1)|C| - 2^{n+1}. \quad (1)$$

Let $S = \{y : g(y) \geq 1\}$ be the *over-covered set* and put $s = |S|$. Since g is non-negative and integer-valued, and vanishes off S ,

$$s \leq E. \quad (2)$$

Finally we use the ball intersection numbers of the cube: for $x \neq z$,

$$|B(x) \cap B(z)| = \begin{cases} 2, & d(x, z) \in \{1, 2\}, \\ 0, & d(x, z) \geq 3, \end{cases} \quad (3)$$

together with $|B(x) \cap B(x)| = n+1$. For $d(x, z) = 1$ the two common points are x and z themselves; for $d(x, z) = 2$ they are the two words lying between them.

3. Proof of Theorem 1.1

Proof. Step 1 (parity). Fix $x \in C$ and sum the coverage function over its ball. Exchanging the order of summation and applying (3),

$$\sum_{y \in B(x)} c(y) = \sum_{z \in C} |B(x) \cap B(z)| = (n+1) + 2N_2(x),$$

where $N_2(x) = \#\{z \in C : 1 \leq d(x, z) \leq 2\}$. The term $n+1$ is the contribution of $z = x$, which lies in C by assumption and, because C is a set, occurs exactly once; every other codeword contributes 0 or 2. Because n is even, $n+1$ is odd, so the whole sum is *odd*. On the other hand it is a sum of $n+1$ terms each at least 2, so it is at least $2(n+1)$; being odd, it is at least $2(n+1) + 1$. Therefore

$$\sum_{y \in B(x)} g(y) \geq 1,$$

so every codeword x admits some $y \in B(x)$ with $g(y) \geq 1$, that is, with $y \in S$.

Step 2 (incidence count). Count the pairs (x, y) with $x \in C$, $y \in S$ and $y \in B(x)$. By Step 1 each $x \in C$ occurs at least once, and each $y \in S$ occurs exactly $|B(y) \cap C| = c(y)$ times. Hence

$$|C| \leq \sum_{y \in S} c(y) = \sum_{y \in S} (g(y) + 2) = E + 2s.$$

Step 3. By (2) we have $s \leq E$, so $|C| \leq 3E$. Substituting (1),

$$|C| \leq 3(n+1)|C| - 3 \cdot 2^{n+1}, \quad \text{that is} \quad 3 \cdot 2^{n+1} \leq (3n+2)|C|. \quad \square$$

Remark 3.1. *The parity step is essential, and the conclusion genuinely fails for odd n . For example, in $\mathbb{F}_2^3$ the code $C = \{000, 001, 110, 111\}$ has $c(x) = 2$ for every x , so it is a perfect twofold covering with $E = 0$ and $|C| = 4 = K(3, 1, 2)$, whereas the formula of Theorem 1.1 would demand $\lceil 48/11 \rceil = 5$.*

Remark 3.2. *The hypothesis that C is a set is essential for this parity argument. If a codeword x could occur with multiplicity m_x , the diagonal contribution in Step 1 would be $m_x(n+1)$, whose parity is not determined when m_x may be even. The incidence count of Step 2 adapts to multiplicities; it is the uniqueness of the diagonal term that fails.*

4. Comparison with the Krotov–Potapov Bound

Krotov and Potapov [2, Theorem 6] give, with $\mu \equiv \tau \pmod{2}$ and $\tau \in \{0, 1\}$,

$$K(n, 1, \mu) \geq \frac{2^n(\mu n + 3\mu + \tau)}{n(n+4)} \quad (n \equiv 0 \pmod{4}), \quad K(n, 1, \mu) \geq \frac{2^n(\mu n + \mu + \tau)}{n(n+2)} \quad (n \equiv 2 \pmod{4}).$$

Write $L(n) = 6 \cdot 2^n / (3n+2)$ for the bound of Theorem 1.1, and $KP(n)$ for the relevant right-hand side above at $\mu = 2$, $\tau = 0$.

Theorem 4.1. *For every even $n \geq 6$ we have $\lceil L(n) \rceil > \lceil KP(n) \rceil$.*

Proof. Cross-multiplying gives the exact identities

$$L - KP = \frac{2^n(2n-12)}{n(n+4)(3n+2)} \quad (n \equiv 0 \pmod{4}), \quad L - KP = \frac{2^n(2n-4)}{n(n+2)(3n+2)} \quad (n \equiv 2 \pmod{4}),$$

so the sign is decided by $2n-12$ and $2n-4$ respectively, and $L > KP$ for every even $n \geq 6$. (At $n=4$ the number KP is the larger of the two, but both ceilings equal 7, so no integer bound is lost.)

The inequality between real numbers does not immediately give the inequality between ceilings, and the gap is genuinely small exactly where it matters: it equals $8/15$ at $n=6$ and $16/39$ at $n=8$, both below 1. Those two cases are therefore checked directly, giving $20 > 19$ and $60 > 59$. At $n=10$ the gap is $64/15 > 1$. For even $n \geq 12$ it suffices to bound the smaller of the two gaps. Since

$$\frac{2n-4}{n(n+2)(3n+2)} \geq \frac{2n-12}{n(n+4)(3n+2)},$$

the residue class $n \equiv 0 \pmod{4}$ is the binding one. Using $2n-12 \geq n/2$ (valid for $n \geq 8$) and $n(n+4)(3n+2) \leq 8n^3$, which is equivalent to $5n^2 - 14n - 8 \geq 0$ and so holds for $n \geq 4$,

$$L - KP \geq \frac{2^n(n/2)}{8n^3} = \frac{2^n}{16n^2} > 1.$$

Indeed $2^{11} = 2048 > 1936 = 16 \cdot 11^2$, and $2^n/n^2$ is strictly increasing for $n \geq 3$ because $(2^{n+1}/(n+1)^2)/(2^n/n^2) = 2n^2/(n+1)^2 > 1$ there. A gap exceeding 1 forces $\lceil L \rceil \geq L > KP + 1 > \lceil KP \rceil$. $\square$

Both bounds are asymptotically $2^{n+1}/n$, so their ratio tends to 1 and the improvement may look negligible. It is not. The *additive* gap is asymptotically $(2/3) \cdot 2^n/n^2$, which is unbounded, so the improvement does not run out; over the even n in $[6, 200]$ it survives the ceiling at all 98 values. Table 4.1 records the tabulated positions.

n	best published lower bound	Theorem 1.1	gain
6	20 (exact value, [3])	20	0, matched without search
8	59 [2]	60	+1
10	188 [2, 3]	192	+4
12	640 [2]	647	+7
14	2195 [2]	2235	+40
16	7783 [2]	7865	+82

Table 4.1: Lower bounds on $K(n, 1, 2)$ at the tabulated even lengths.

The entries for $n \geq 8$ are exactly the $\mu = 2$ positions printed in [2], and each printed lower bound there equals the value of their formula, so no tabulated entry is stronger than the formula itself.

5. Consequences at Small n

The value $K(6, 1, 2) = 20$. Theorem 1.1 gives $\lceil 384/20 \rceil = 20$. Combined with the known twenty-word twofold covering of $\mathbb{F}_2^6$, recorded explicitly in [4] as

$$\{0, 1, 2, 4, 8, 23, 27, 29, 30, 31, 39, 43, 45, 46, 47, 48, 49, 50, 52, 56\}$$

(written as integers in $[0, 63]$), this recovers $K(6, 1, 2) = 20$. The value was previously established computationally [3]; Theorem 1.1 replaces the lower-bound half of that determination by a one-line argument, while the upper half still rests on the construction above.

The bound $K(8, 1, 2) \geq 60$. Theorem 1.1 gives $\lceil 1536/26 \rceil = 60$, one above the published 59. The sequence A004045 in the On-Line Encyclopedia of Integer Sequences [4] currently lists exact values only through $n = 7$, namely 2, 3, 4, 8, 12, 20, 32, so $n = 8$ is its first unresolved position. Krotov and Potapov record $59 \leq K(8, 1, 2) \leq 64$ [2], the upper bound being the doubling $K(8, 1, 2) \leq 2K(7, 1, 2) = 64$ already present in the table of [1]. Theorem 1.1 raises the lower endpoint, giving $60 \leq K(8, 1, 2) \leq 64$.

6. Even Multiplicities

Carry a general μ through the proof. For even n the ball sum $(n+1) + 2N_2(x)$ at a codeword is odd, so parity alone forces it to exceed the baseline $\mu(n+1)$ exactly when $\mu(n+1)$ is even — that is, exactly when μ is even.

Proposition 6.1. *Let $n \geq 2$ be even and let μ be an even integer with $2 \leq \mu \leq n$. (Since $|B(x)| = n + 1$ and C is a set, no μ -fold covering exists for $\mu > n + 1$, so $\mu = n$ is the largest feasible even multiplicity.) Then*

$$K(n, 1, \mu) \geq \frac{\mu(\mu + 1)2^n}{(\mu + 1)(n + 1) - 1}, \quad \text{and hence} \quad K(n, 1, \mu) \geq \left\lceil \frac{\mu(\mu + 1)2^n}{(\mu + 1)(n + 1) - 1} \right\rceil.$$

Proof. Put $g_\mu(y) = c(y) - \mu$ and $E_\mu = \sum_y g_\mu(y) = (n + 1)|C| - \mu 2^n$, and let $S_\mu = \{y : g_\mu(y) \geq 1\}$ with $s_\mu = |S_\mu| \leq E_\mu$. For n and μ both even, $\mu(n + 1)$ is even while the ball sum $(n + 1) + 2N_2(x)$ at a codeword is odd, so that sum exceeds $\mu(n + 1)$ and each codeword ball meets S_μ . Counting incidences as in Step 2,

$$|C| \leq \sum_{y \in S_\mu} c(y) = E_\mu + \mu s_\mu \leq (\mu + 1)E_\mu = (\mu + 1)((n + 1)|C| - \mu 2^n),$$

which rearranges to the stated bound. $\square$

At $\mu = 2$ this is Theorem 1.1. For larger even μ the comparison with [2] depends on both parameters. Writing $P(n, \mu)$ for the right-hand side of Proposition 6.1 and $D = (\mu + 1)(n + 1) - 1$, the exact differences are

$$P - KP = \frac{\mu 2^n (n - 3\mu)}{D n (n + 4)} \quad (n \equiv 0 \pmod{4}), \quad P - KP = \frac{\mu 2^n (n - \mu)}{D n (n + 2)} \quad (n \equiv 2 \pmod{4}),$$

so that, as a comparison of the underlying real-valued bounds,

$$P > KP \iff \mu < n/3 \quad (n \equiv 0 \pmod{4}), \quad P > KP \iff \mu < n \quad (n \equiv 2 \pmod{4}).$$

The gain is therefore not confined to $\mu = 2$. At $\mu = 4$, for instance, Proposition 6.1 gives 380 at $n = 10$ and 15604 at $n = 16$, against 376 and 15565 from [2]; it loses at $n = 8$, where it gives 117 against 118. As in Section 4, a strict inequality between the real bounds need not survive the ceiling: at $n = 6$, $\mu = 4$ the difference is $16/51 > 0$, yet both ceilings equal 38. We therefore state the classification for the real-valued bounds only, and do not claim a corresponding classification after rounding.

7. Relation to Previous Work

The covering excess method for *ordinary* coverings goes back to van Wee [5], whose radius-one bound is asymptotically $2^n/n$; that work concerns $K(n, R)$ and does not yield the present bound. The extension of excess counting to multiple coverings is [1]. In their Theorem 6 the correction term at radius one is

$$\varepsilon = 2 \left\lceil \frac{\mu(n + 1)}{2} \right\rceil - \mu(n + 1),$$

so $\varepsilon = 0$ exactly when $\mu(n + 1)$ is even. For even n the integer $n + 1$ is odd, so this happens exactly for even μ , and their bound then reduces to the sphere covering bound $K(n, 1, \mu) \geq \mu 2^n / (n + 1)$. The present argument instead uses the parity of the ball sum at a codeword, which is odd whenever n is even for every μ , and therefore applies precisely where ε vanishes.

Reference [1] also cites a forthcoming manuscript of W. Chen and D. Li entitled *Lower bounds for multiple covering codes*. The author has not located a published version; the novelty comparison above is therefore limited to the accessible published literature.

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